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BlockChain Lab

Report
on

Digital Evidence Chain-of-Custody Tracker

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Abstract

In the modern digital era, evidence in investigations increasingly consists of digital data such as images, videos, and documents. Ensuring the integrity, authenticity, and traceability of such evidence is a critical challenge, as traditional systems rely on centralized databases that are susceptible to tampering, unauthorized access, and data manipulation. Any compromise in the chain-of-custody can lead to weakened legal cases and loss of trust in the judicial process. This project proposes a Digital Evidence Chain-of-Custody Tracker that utilizes blockchain technology to provide a secure, transparent, and tamper-proof mechanism for managing evidence. The system is integrated with an existing AI-based application, LookBack, which captures and categorizes evidence and stores metadata in Firebase. A cryptographic hash of each evidence file is generated using SHA-256 and stored on a blockchain through smart contracts deployed on a local Ethereum network using Ganache. This hybrid architecture ensures efficient storage and retrieval through Firebase while leveraging blockchain for immutable verification. The proposed system enhances reliability, accountability, and trust in evidence handling, making it highly suitable for applications in law enforcement, digital forensics, and legal proceedings.

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Chapter 1

Introduction

1.1 Introduction

Digital transformation has significantly impacted investigative processes, with digital evidence playing a vital role in criminal and civil cases. Evidence such as photographs, videos, and electronic documents must be handled carefully to ensure its authenticity and integrity. One of the most critical aspects of evidence management is maintaining a proper chain-of-custody, which records every transfer and handling of evidence. Traditional evidence management systems rely on centralized databases and manual logging mechanisms. These systems are vulnerable to tampering, unauthorized modifications, and lack transparency. Even a minor inconsistency in the chain-of-custody can result in evidence being deemed inadmissible in court. Blockchain technology offers a decentralized and immutable solution to this problem. By storing evidence records in a blockchain, it becomes nearly impossible to alter past data without detection.

1.2 Literature Review

Table 1: Literature Review

Title & Authors	Aim of the Paper	Limitations Identified
[1] *Blockchain-Based Chain of Custody for Digital Forensics* – M. Conti, A. Dehghantanha, K. R. Choo	To provide a secure and tamper-proof mechanism for tracking digital evidence using blockchain in forensic investigations	Primarily conceptual; lacks real-time implementation and integration with existing systems
[2] *A Blockchain-Based Framework for Secure Evidence Management* – Z. Zhang, X. Chen	To develop a decentralized system for storing and verifying digital evidence securely	High storage cost when storing large data on-chain; performance issues in large-scale systems
[3]*Blockchain for Digital Forensics: Opportunities and Challenges* – M. Conti, S. Kumar Karuppayah	To analyze how blockchain can improve reliability and transparency in digital forensics	Focuses on theoretical analysis; lacks practical deployment and performance evaluation

1.3 Problem Definition

In current digital evidence management systems, data is typically stored in centralized databases, making it vulnerable to tampering, unauthorized access, and accidental modification. The lack of a secure and transparent mechanism for maintaining the chain-of-custody can compromise the integrity of evidence. This can lead to legal challenges, reduced credibility of evidence, and potential failure of investigations. Therefore, there is a need for a system that ensures secure, tamper-proof, and verifiable handling of digital evidence throughout its lifecycle.

1.4 Objectives

1. To design a secure system for managing digital evidence
2. To generate cryptographic hashes for evidence files using SHA-256
3. To store evidence hashes on blockchain for immutability
4. To maintain a transparent and verifiable chain-of-custody
5. To integrate blockchain with Firebase for efficient data handling
6. To enable verification of evidence integrity
7. To demonstrate a practical real-world application of blockchain

1.4 Proposed Solution

1. The proposed system introduces a hybrid architecture that combines the efficiency of centralized databases with the security of blockchain technology. The LookBack application captures and categorizes evidence using AI, storing metadata and files in Firebase.
2. A cryptographic hash of each evidence file is generated using SHA-256 and stored on a blockchain through a smart contract. This ensures that any modification to the evidence can be detected by comparing the current hash with the stored hash.
3. The blockchain maintains a permanent and immutable record of evidence creation and transfer, enabling a reliable chain-of-custody. QR codes can be generated to link physical evidence with its corresponding digital record. This approach ensures both efficiency and security in evidence management.

1.4 Technology / Platform Used

1. Solidity – Used to develop smart contracts
2. Ethereum (Ganache) – Local blockchain network for testing
3. Remix IDE – Smart contract development and deployment
4. MetaMask – Wallet for authentication and transactions
5. Node.js – Backend communication with blockchain
6. Firebase – Storage of metadata and images
7. SHA-256 Algorithm – Hash generation for integrity
8. Ethers.js – Blockchain interaction library

Chapter 2

System Description

2.1 System Block Diagram

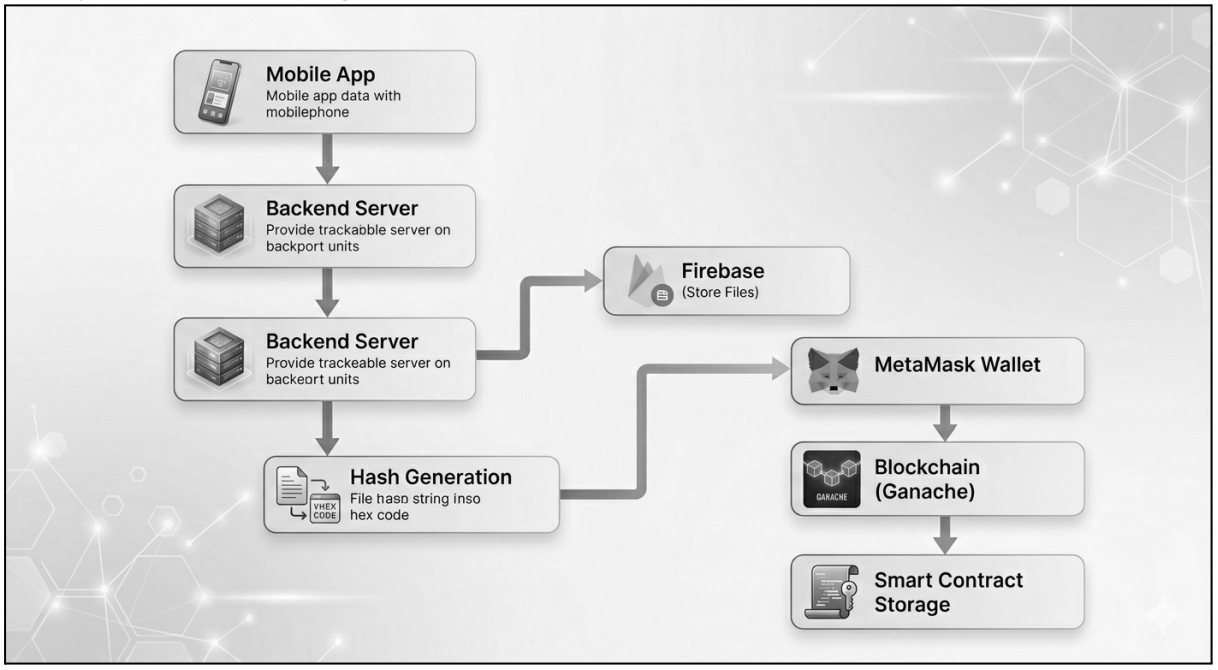

Figure 1: Architecture Diagram

The system begins with the mobile application capturing evidence data, which is sent to the backend server for processing and storage. The media files are stored in Firebase for efficient access, while a unique SHA-256 hash is generated for each file. This hash is then sent through MetaMask to the blockchain (Ganache), where it is stored using a smart contract. This ensures that any tampering with the original file can be detected, providing a secure and verifiable chain of evidence.

2.2 Software Requirements

Table 2: Software Requirements

Category	Requirements
Platform & Tools	Windows/Linux/macOS, Google Chrome, Ganache, MetaMask, Remix IDE
Development	Solidity, JavaScript/TypeScript, Node.js
Database & Libraries	Firebase Firestore, Ethers.js

2.3 Flowchart

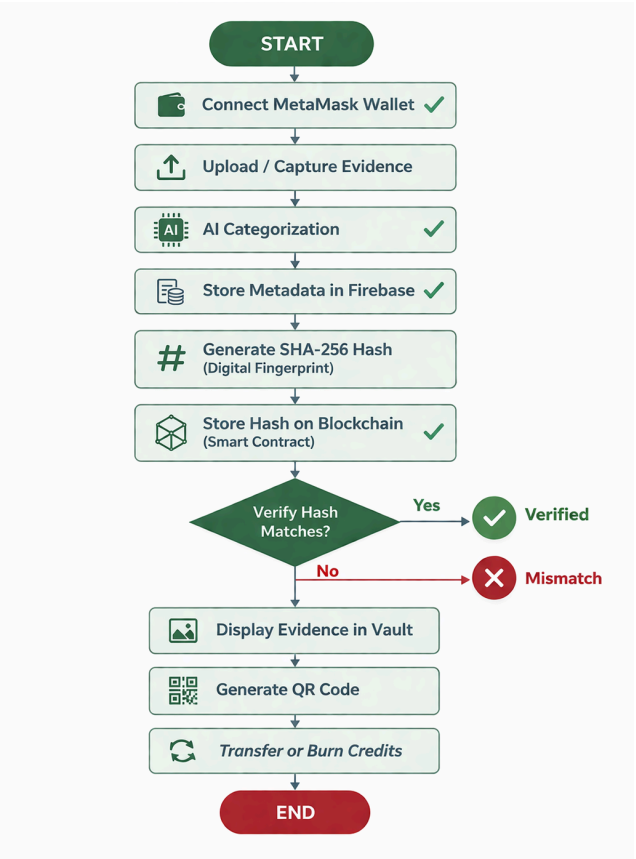

Figure 2: Flowchart

The flowchart represents the working of the LookBack Blockchain Evidence System. It begins with connecting the MetaMask wallet, followed by capturing or uploading evidence. The system processes the evidence using AI and stores its metadata in Firebase. A SHA-256 hash of the evidence is generated and stored on the blockchain through a smart contract. The system then verifies the integrity of the evidence by comparing hashes. Based on the result, the evidence is marked as verified or mismatched, and finally displayed in the evidence vault for further actions.

Chapter 3

Results and Discussions

3.1 Code

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract EvidenceStorage {
    struct EvidenceRecord {
        string evidenceId;
        string evidenceHash;
        uint256 timestamp;
        address officerAddress;
    }
    mapping(string => EvidenceRecord) public evidenceRecords;
    mapping(string => bool) public evidenceExists;

    event EvidenceStored(
        string evidenceId,
        string evidenceHash,
        uint256 timestamp,
        address officerAddress
    );
    function createEvidence(string memory _evidenceId, string memory _evidenceHash)
public {
        require(!evidenceExists[_evidenceId], "Evidence already exists");

        evidenceRecords[_evidenceId] = EvidenceRecord({
            evidenceId: _evidenceId,
            evidenceHash: _evidenceHash,
            timestamp: block.timestamp,
            officerAddress: msg.sender
        });
        evidenceExists[_evidenceId] = true;

        emit EvidenceStored(_evidenceId, _evidenceHash, block.timestamp, msg.sender);
    }
    function getEvidence(string memory _evidenceId) public view returns (
        string memory,
        string memory,
        uint256,
        address
```

```
    ) {
        require(evidenceExists[_evidenceId], "Evidence does not exist");

        EvidenceRecord memory record = evidenceRecords[_evidenceId];
        return (
            record.evidenceId,
            record.evidenceHash,
            record.timestamp,
            record.officerAddress
        );
    }
    function verifyEvidence(string memory _evidenceId, string memory _hashToVerify)
public view returns (bool) {
        if (!evidenceExists[_evidenceId]) {
            return false;
        }
        EvidenceRecord memory record = evidenceRecords[_evidenceId];
        return keccak256(abi.encodePacked(record.evidenceHash)) ==
keccak256(abi.encodePacked(_hashToVerify));
    }
}
```

3.2 Implementation

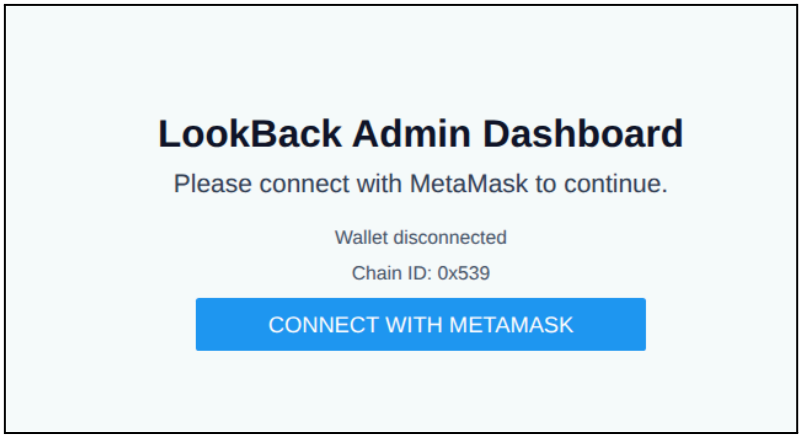

Figure 3: Metamask Connect

This screen represents the LookBack admin dashboard, where the user connects their MetaMask wallet to interact with the blockchain system. It displays the wallet connection status and the current network (Ganache). This step is essential to authenticate the user and ensure that all blockchain operations, such as storing and verifying evidence, are performed on the correct network.

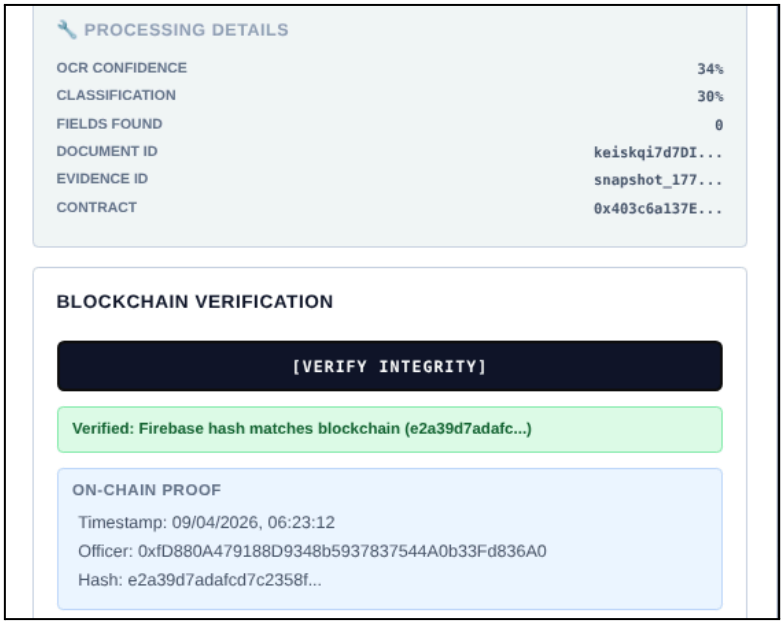

Figure 4: App Verification & Processing

This section shows the AI processing details and verification result. The system processes the captured data (OCR, classification, etc.) and compares the current file hash with the stored blockchain hash. The "Verified" message confirms that the evidence is authentic and untampered, maintaining integrity and trust.

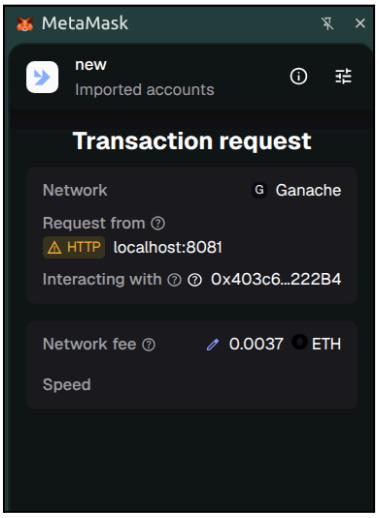

Figure 5: MetaMask Transaction

This shows that a blockchain transaction is being initiated using MetaMask. The system sends the SHA-256 hash of the evidence to the blockchain (Ganache) through a smart contract. The transaction confirms that the hash is securely recorded on-chain, ensuring the evidence cannot be altered without detection.

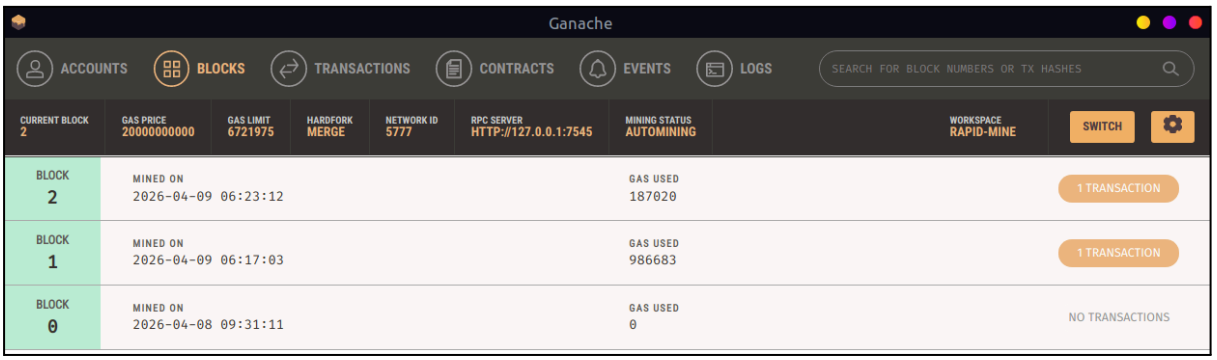

Figure 6: Ganache blockchain

This panel displays the Ganache blockchain blocks, where each block represents a confirmed transaction. It shows that the evidence hash has been successfully added to the blockchain, creating a permanent and immutable record. This ensures transparency and traceability of evidence storage.

3.3 Future Directions

- Integration with permissioned blockchains like Hyperledger Fabric for real-world institutional use
- Implementation of multi-signature authentication for secure evidence transfers
- Use of IPFS (InterPlanetary File System) for decentralized evidence storage
- Development of a mobile application for real-time evidence capture and verification
- Integration with law enforcement and legal systems for practical deployment
- Enhancement using AI-based tamper detection and anomaly detection
- Deployment on public or scalable networks (e.g., Polygon) for wider accessibility

Chapter 4

Conclusion

The LookBack system provides a comprehensive and scalable solution for enhancing evidence collection and management in modern law enforcement and security operations. By integrating IoT-enabled smart glasses with AI-based processing and blockchain-backed verification, the system enables hands-free data capture, real-time monitoring, and intelligent organization of evidence. The incorporation of technologies such as OCR for text extraction, object detection for visual tagging, and machine learning models for document classification significantly improves the accuracy, efficiency, and accessibility of collected data. Additionally, the use of SHA-256 hashing and blockchain storage ensures data integrity, transparency, and tamper-proof evidence handling, which is critical in legal and investigative scenarios. The modular architecture of the system allows seamless integration of additional features and ensures flexibility for future upgrades. Overall, LookBack reduces reliance on manual processes, enhances situational awareness for field personnel, and establishes a reliable and secure framework for digital evidence management, making it highly suitable for real-world deployment in security and surveillance applications.

References

[1] M. Conti, A. Dehghantanha, and K. R. Choo, “Blockchain-based chain of custody for digital forensics,” IEEE Transactions on Information Forensics and Security, vol. 15, pp. 1–12, 2020.

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[4] H. Tian, J. He, and Y. Ding, “A secure data sharing system based on Hyperledger Fabric blockchain,” Future Generation Computer Systems, vol. 95, pp. 88–98, 2019.

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